RESULT 1 / VALID, WITH FEWER ERRORS

Every ontology is valid. RIGOR's have far fewer modeling errors.

All 30 ontologies are valid OWL 2 DL (RDFLib) and logically consistent (HermiT). So we look deeper.

MODELING ERRORS FOUND BY OOPS!

pitfalls per ontology, fewer is better

	Liver Cancer			eICU-CRD			Chinook			Avg
	Claude	Mistral	DeepSeek	Claude	Mistral	DeepSeek	Claude	Mistral	DeepSeek	
Direct Mapping	731			665			102			499
LLM baseline	422	397	403	434	440	421	95	74	81	307
Non-Iterative	29	61	18	39	220	27	24	35	14	52
RIGOR	**20**	**23**	**3**	**2**	**2**	**26**	**0**	**0**	**0**	**8**

Same LLM, same database: **Mistral on eICU, 220 without RIGOR, 2 with it.** On Chinook, RIGOR has zero with every LLM.

Validity: RDFLib; consistency: HermiT (paper Sec. 4.7). Pitfalls: OOPS!, paper Table 7; 72% of the other methods' pitfalls are missing annotations (P08).

RESULT 2 / NOTHING IS LOST

Nothing is lost: RIGOR keeps the schema.

Share of database columns covered, for every LLM and database.

1 VALID, FEWER ERRORS | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | 4 GOLD MAPPINGS | 5 JUDGES

KEEPS THE SCHEMA

columns covered, per ontology (%)

	Liver Cancer			eICU-CRD			Chinook			Avg
	Claude	Mistral	DeepSeek	Claude	Mistral	DeepSeek	Claude	Mistral	DeepSeek	
Direct Mapping	97			99			78			91%
LLM baseline	94	31	89	96	68	86	100	50	52	74%
Non-Iterative	41	6	12	22	28	46	58	16	78	34%
RIGOR	**91**	**99**	**87**	**100**	**99**	**77**	**95**	**95**	**95**	**93%**

Same context, no seed: **Non-Iterative 6–78%**, depending on the LLM. With RIGOR: **≥ 77% for every LLM, in every database.**

Coverage: a column counts as covered when an ontology class or property name is similar to it (embedding similarity ≥ 0.55). Paper Sec. 4.6, Table 9.

RESULT 3 / MEANING IS ADDED

Meaning is added: labels, provenance and hierarchy.

Per ontology, for every LLM and database.

1 VALID, FEWER ERRORS | 2 KEEPS THE SCHEMA | **3 ADDS MEANING** | 4 GOLD MAPPINGS | 5 JUDGES

ADDS MEANING

		Liver Cancer			eICU-CRD			Chinook			Avg
		Claude	Mistral	DeepSeek	Claude	Mistral	DeepSeek	Claude	Mistral	DeepSeek	
	Direct Mapping LLM baseline	*No labels, provenance or class hierarchy, in any run.*									
Labels & comments	Non-Iterative RAG	680	134	736	877	157	743	98	14	64	389
	RIGOR	**1,029**	**1,137**	**341**	**1,291**	**1,684**	**435**	**357**	**362**	**359**	**777**
Provenance links	Non-Iterative RAG	0	0	0	0	0	0	0	0	0	0
	RIGOR	**336**	**308**	**22**	**423**	**335**	**99**	**81**	**83**	**75**	**196**
Class hierarchy *inheritance richness*	Non-Iterative RAG	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	RIGOR	**0.70**	**0.39**	**0.13**	**0.62**	**0.49**	**0.10**	**2.12**	**2.72**	**4.07**	**1.26**

Only RIGOR adds labels, provenance **and** a class hierarchy: **with every LLM, in every database.**

Labels & comments: annotation assertions; provenance: prov:wasDerivedFrom (paper Table 8). Hierarchy: inheritance richness, subclass axioms per class [OntoQA: Tartir et al., KADASH 2005; OntoMetrics: Lantow, KEOD 2016], from Table 8. Shading compares cells within each database.

RESULT 4 / EXTERNAL BENCHMARK

RIGOR keeps the link to the data.

BURR [Laskowski et al., PACMMOD 2025] · ISWC scenario · highest F1 among ontology-learning systems

1 VALID, FEWER ERRORS | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | **4 GOLD MAPPINGS** | 5 JUDGES

WHAT BURR CHECKS

Every ontology element must map back to the database.

CLASSES	`person`	→	`Class: Person`
RELATIONS	`paper.author_id`	→	`ObjectProperty: author`
ATTRIBUTES	`person.name`	→	`DataProperty: name`

Elements match when they map to the same database elements, whatever their names. F1 per element type.

Illustrative elements · ISWC gold mapping: D2RQ example by Bizer & Seaborne

SYSTEM	CLASSES	RELATIONS	ATTRIBUTES
W3C-Mapper / Ontop rule-based	0.53	0.11	0.39
RDB2Onto rule-based	0.67	0.62	0.44
OntoGenix LLM-based	0.58	0.35	0.44
RIGOR LLM + rules	**0.73**	**0.67**	**0.52**

Relations: 0.67 vs 0.11 for W3C-Mapper / Ontop. Foreign keys become named relations, like hasPatientData.

One scenario · 9 tables · 46 columns · 11 FKs · RIGOR with Claude · its prov:wasDerivedFrom links exported as a D2RQ mapping · Source: paper Table 13

RESULT 5 / LLM JUDGE AND HUMAN EXPERTS

An LLM judge and human experts both rank RIGOR first.

Both ranked four anonymized ontologies per table, judged against that table's competency questions.

1 VALID, FEWER ERRORS | 2 KEEPS THE SCHEMA | 3 ADDS MEANING | 4 GOLD MAPPINGS | **5 JUDGES**

COMPETENCY QUESTIONS what the ontology must answer. 5 per table, 290 in total, expert-verified. *e.g. "Which patients had abnormal physical exam findings?"*

LLM-AS-A-JUDGE · GPT-5.4

58 / 58 tables with RIGOR ranked first

	Liver Cancer	eICU-CRD	Chinook
Direct Mapping	2.05	2.13	0.13
LLM baseline	1.38	0.28	1.48
Non-Iterative	1.89	0.67	2.77
RIGOR	**3.93**	**3.64**	**3.32**

10 ONTOLOGY EXPERTS

28 / 40 rankings with RIGOR first (70%)

	Liver Cancer	eICU-CRD
Direct Mapping	3	2
LLM baseline	1	4
Non-Iterative	2	0
RIGOR	**14**	**14**

Direct Mapping: structure without meaning. LLMs alone: meaning without structure. **RIGOR: both, and checked.**

Judge: average score 0–5 per database, Claude ontologies, 58 tables. Experts: first places of 20 per database, 4 medical tables, Kendall's W 0.39–0.72. Paper Tables 1, 6, 12.